Advanced Data Engineering
01: Overview

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Data (engineering) definitions

- **Data**: A representation of **facts**, **concepts** or **instructions** in a formalised manner suitable for **communication**, **interpretation**, or **processing** by humans or by automatic means." (Hicks [1993: 668] quoted by Checkland and Holwell [1998])
- **Data**: An individual fact represented in forms such as text, numbers, graphics, images, sounds, or videos that are out of context and lack inherent meaning (DAMA: 2021)

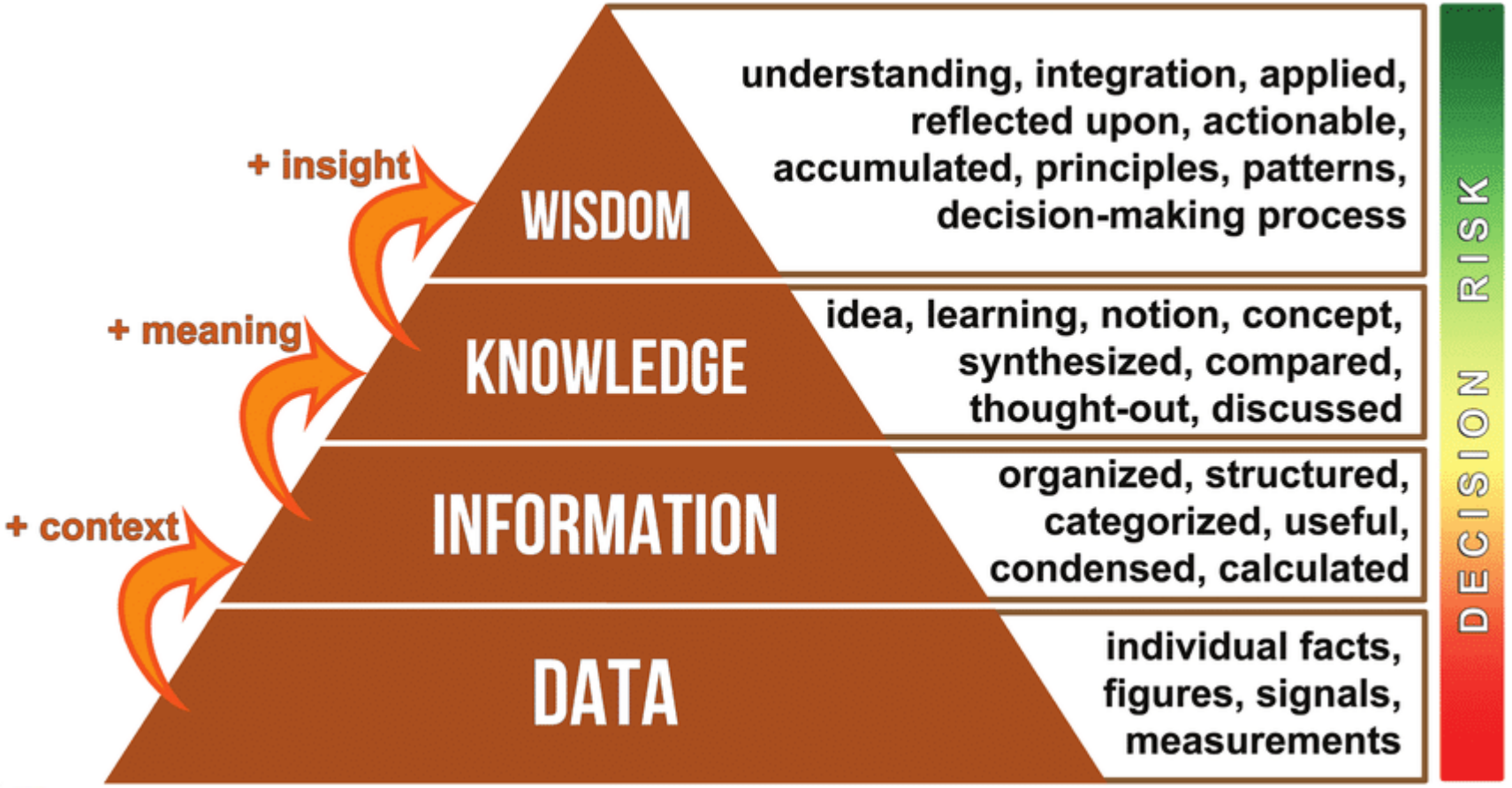
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Software and data engineering

- **Data engineering** is a subdiscipline of software engineering that focuses on the transportation, transformation, and management of data.
- IEEE defines software engineering as "*The application of a **systematic**, **disciplined**, **quantifiable** approach to the **development**, **operation** and **maintenance** of software*."
- **Data engineering**: The application … of software that focuses on the …. of data.

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DIKW Pyramid



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Data is new oil

- Oil has fueled the **industrial economy**.
- The **wealthiest** were those with most oil.
- Oil must be **extracted and refined** into usable products.
- Oil is **finite**.

- Data powers the **information economy**.
- The amount of **knowledge you get** is proportional to data that you have.
- To extract information from data, **collection, processing, and analysis** are required.

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Data for information & knowledge economy

- Amazon - Transforming E-Commerce
  - Personalized Recommendations
  - Anticipatory Shipping
  - Price Setting
- Netflix
  - Move to Big Data: from DVD Rental to a Streaming Service
  - Netflix Contest from 2006 to 2009, 1 Million USD Offer
- Facebook (and other social media) Advertising
  - Realtime tracking of user behavior
  - Movie – The Social Dilemma

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Data solutions



Retail

- Demand Prediction
- Inventory Planning
- Shelf Optimization
- Personalized Content
- Personalized Marketing



Transportation

- Optimized Delivery Scheduling
- Fuel Efficiency Analysis – Airplanes
- Predictive Maintenance
- Congestion Prediction / Management



Banking & Finance

- Automatic Loan Approvals
- Predict New Branch Locations
- Fraud Detection
- Trade Surveillance



Tele-Communication

- Churn Prediction
- Up sell/ Cross Sell Products
- Geo Mapping
- New Product Development
- Price Optimizing

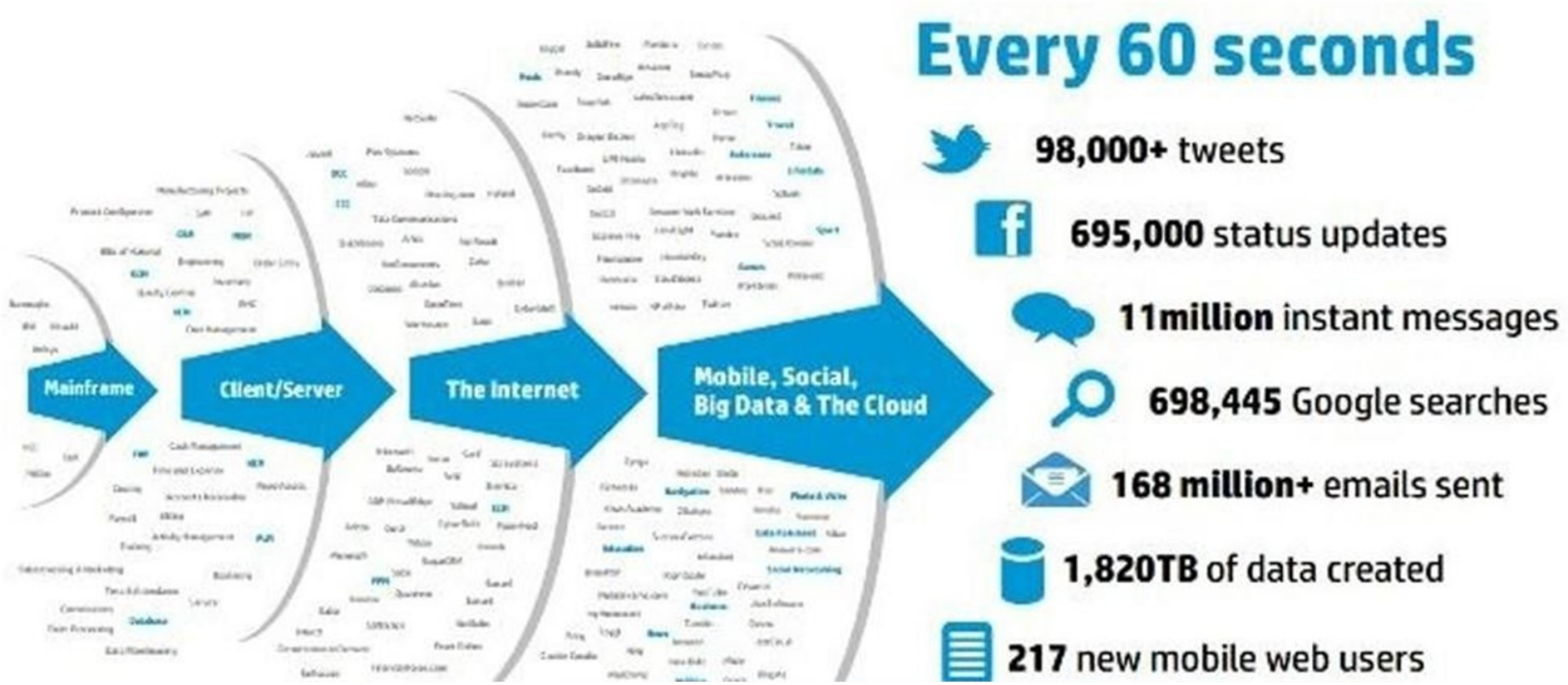


HealthCare

- Disease Diagnosis
- Treatment Recommendations
- Realtime Analytics via wearable devices
- Pandemics – outbreak warnings

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Data never sleeps



Source

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Big data

There are nearly as many pieces of digital information as there are stars in the universe.

175

Zettabytes of data world wide by 2025

$ 140.9 billion

Data science platform market growth by 2024

$127 billion

AI-Based Self-Driving Car Market Is Expected to Reach

12%

Ever used and analyzed

1000

Computers used by Google to search a single query

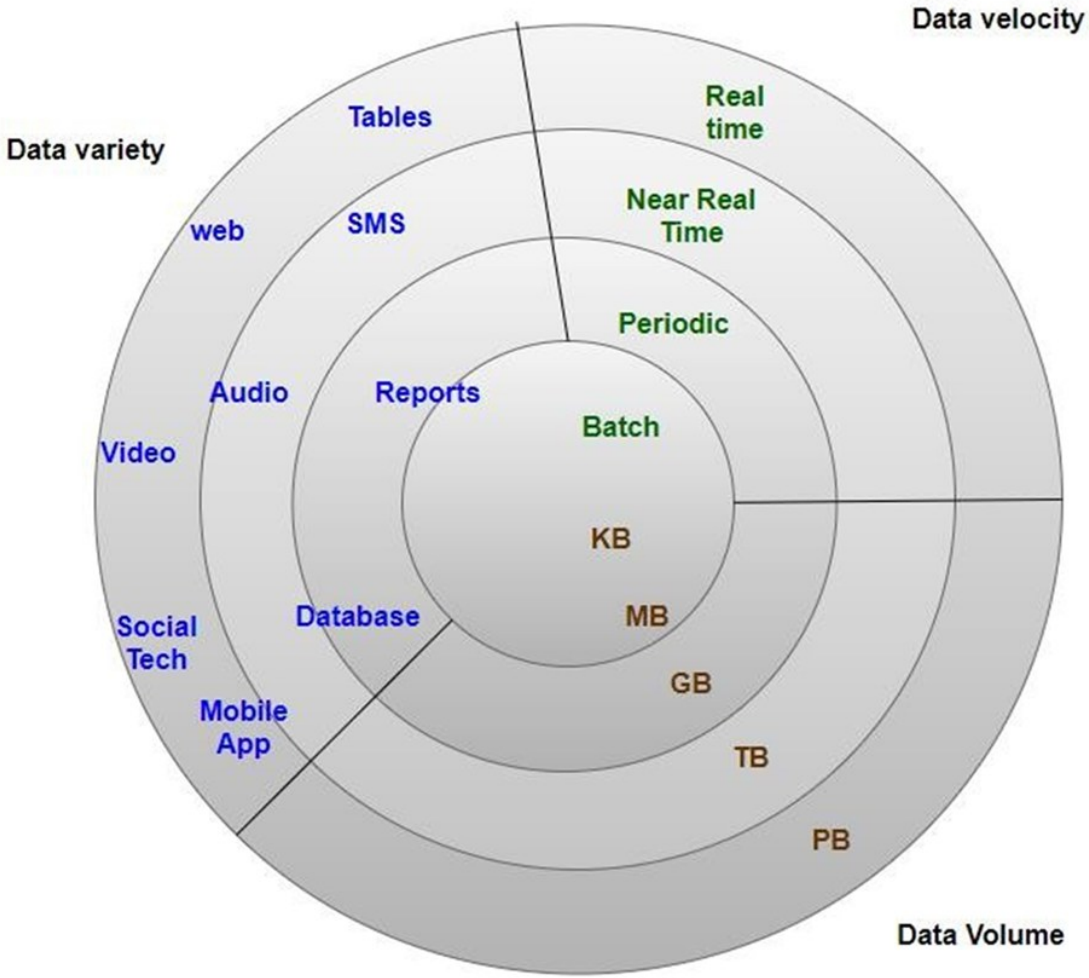
151,717

Nationally, Shortage of people with data related skills in U.S

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Big data characteristics

Big Data is a phrase used to mean a **massive volume** of both **structured and unstructured data** that is so large and which **moves so fast**.



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Big data characteristics

- The V's of Big Data
  - **Volume** – Enough data for the requirement
  - **Velocity** – Speed at which data comes in
  - **Variety** – heterogeneous data sources
  - **Variability** – Data from same source varies with time
  - **Veracity** – Accuracy or truthfulness of a data set
  - **Value** – Access to the data when most needed for business
  - **Visualization** – Can be processed into an understandable format
  - **Vulnerability** – Security of the source

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Data engineering

- Requirements:
  - Need to handle big data in production environments with required speed and accuracy.
  - Responsible for making quality data available for different stakeholders
  - Incomes the data pipelines to serve this purpose.
  - Eliminating the manual steps and automating the process
  - Software Engineers in data driven companies needed to develop tools to handle big data
- Engineering challenges with big data:
  - How to accumulate data from multiple sources?
  - How to store, move and process large volumes of data?
  - How to handle unstructured data?
  - How to find the insights from the huge data?
  - How to filter the required data?
  - How to prevent data loss?

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Data engineer

- Around 2011 the term “Data Engineer” started to appear.

Engineers design and build things. “Data” engineers design, build and maintain pipelines that transform and transport data in a usable format to be used by entire organization.

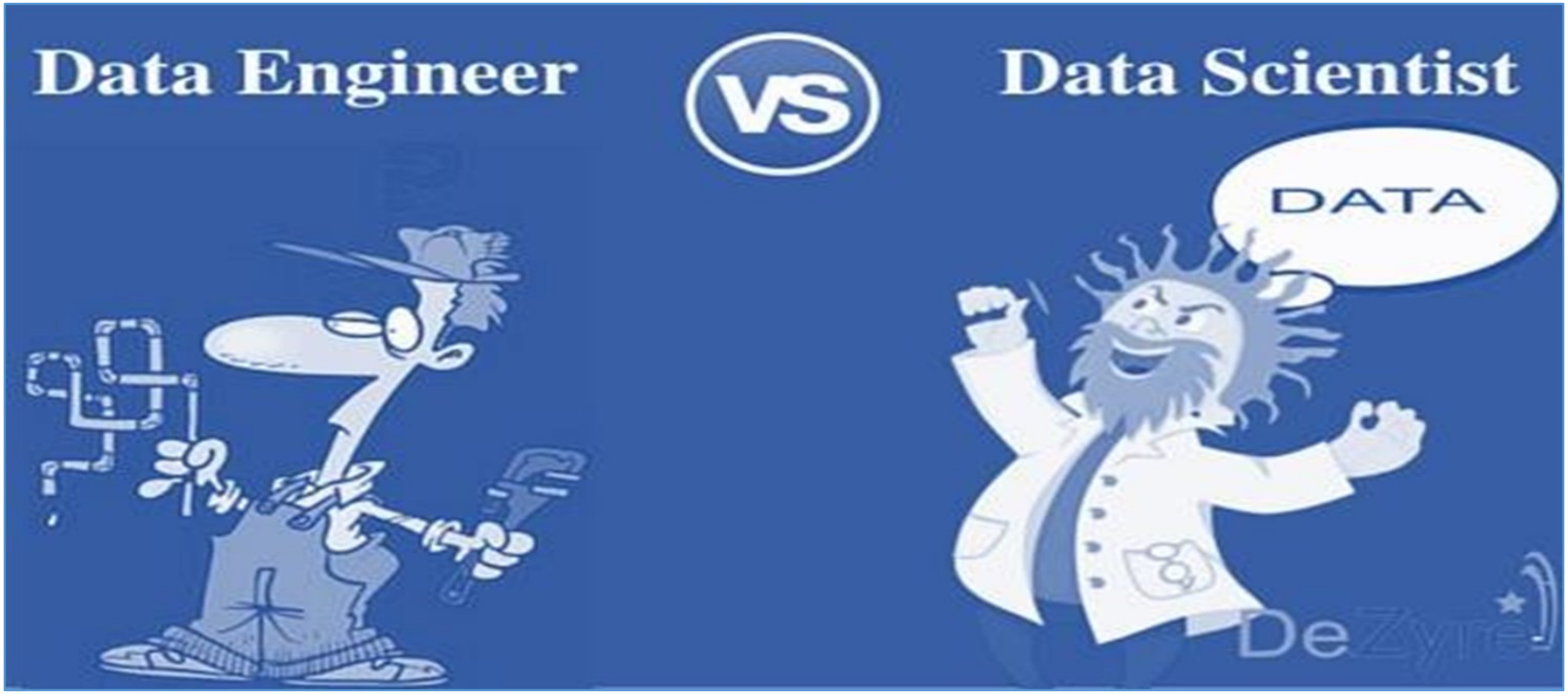
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Data engineer vs the others

- Build and Optimize systems and pipelines handling large volumes of data. Make clean and quality data accessible for anyone who needs it.
- Vs. Data Scientist
  - Deals with different types of data and different approaches to bring value out of data
  - Make use of loaded data (e.g., in a data warehouse) to experiment and produce best models for the business requirement
- Vs. Machine Learning Engineer
  - Take Data Science models into Production
  - Design and implement scheduled processes for re-training and batch scoring

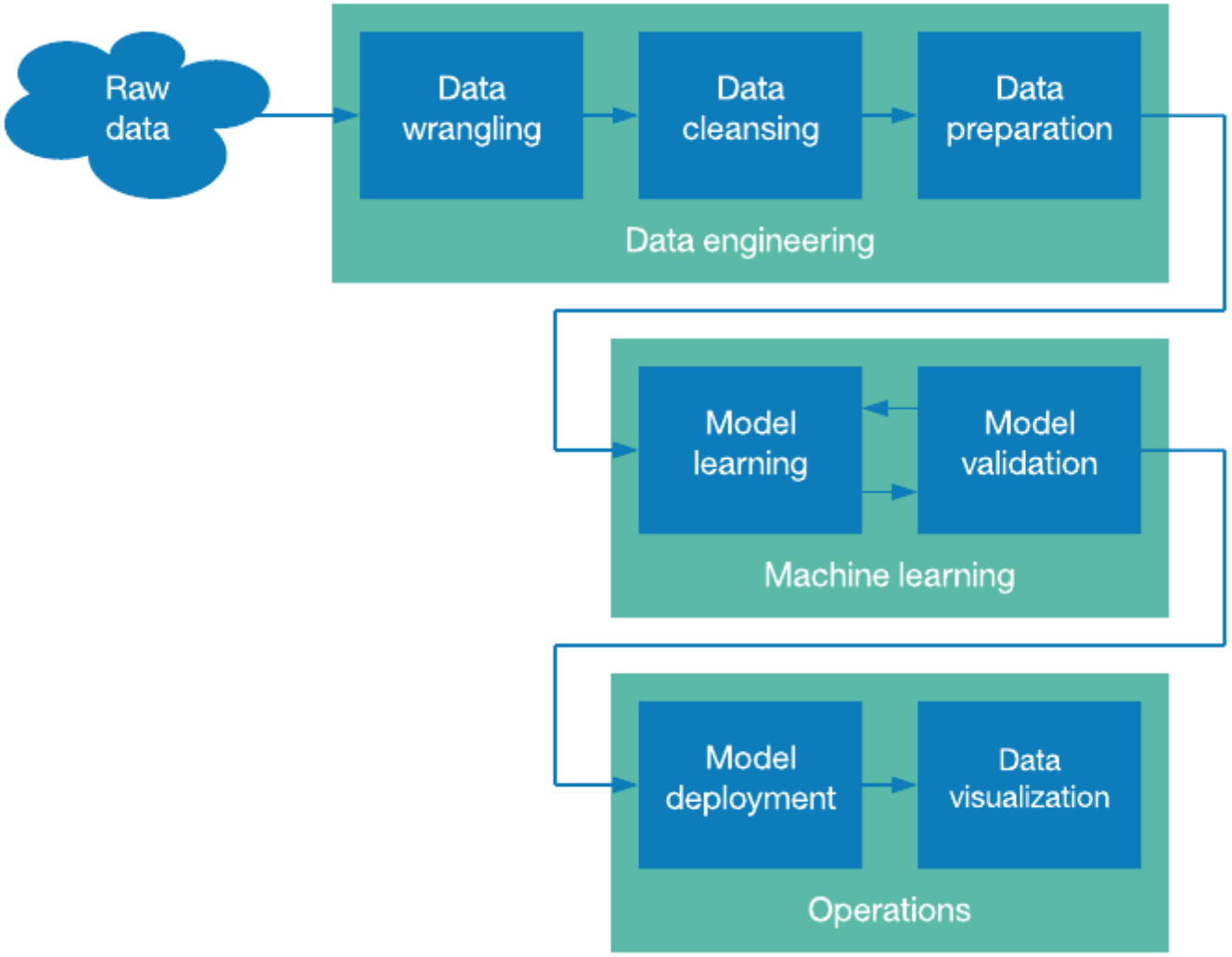
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Data engineer vs the others



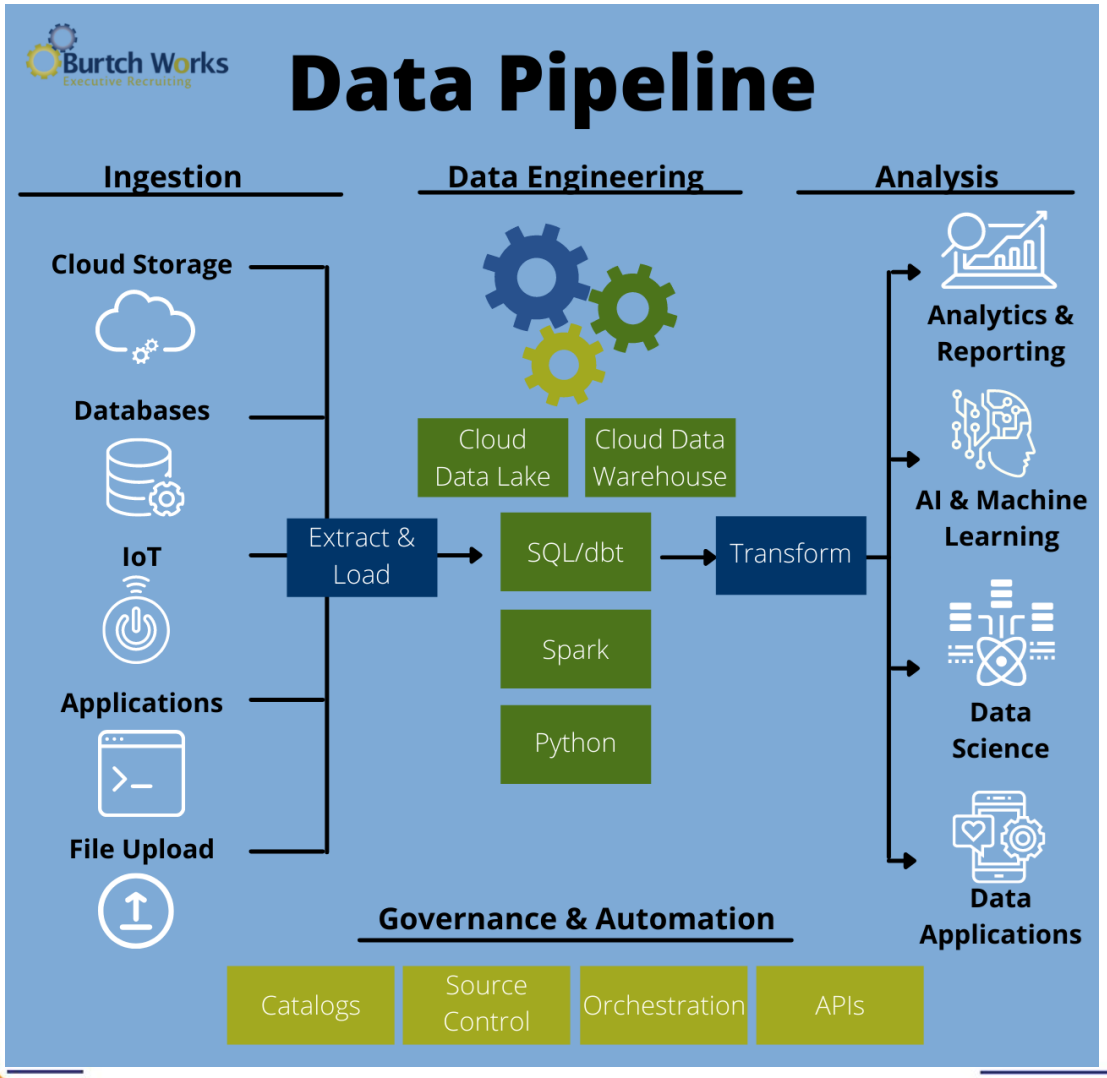
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Data pipelines



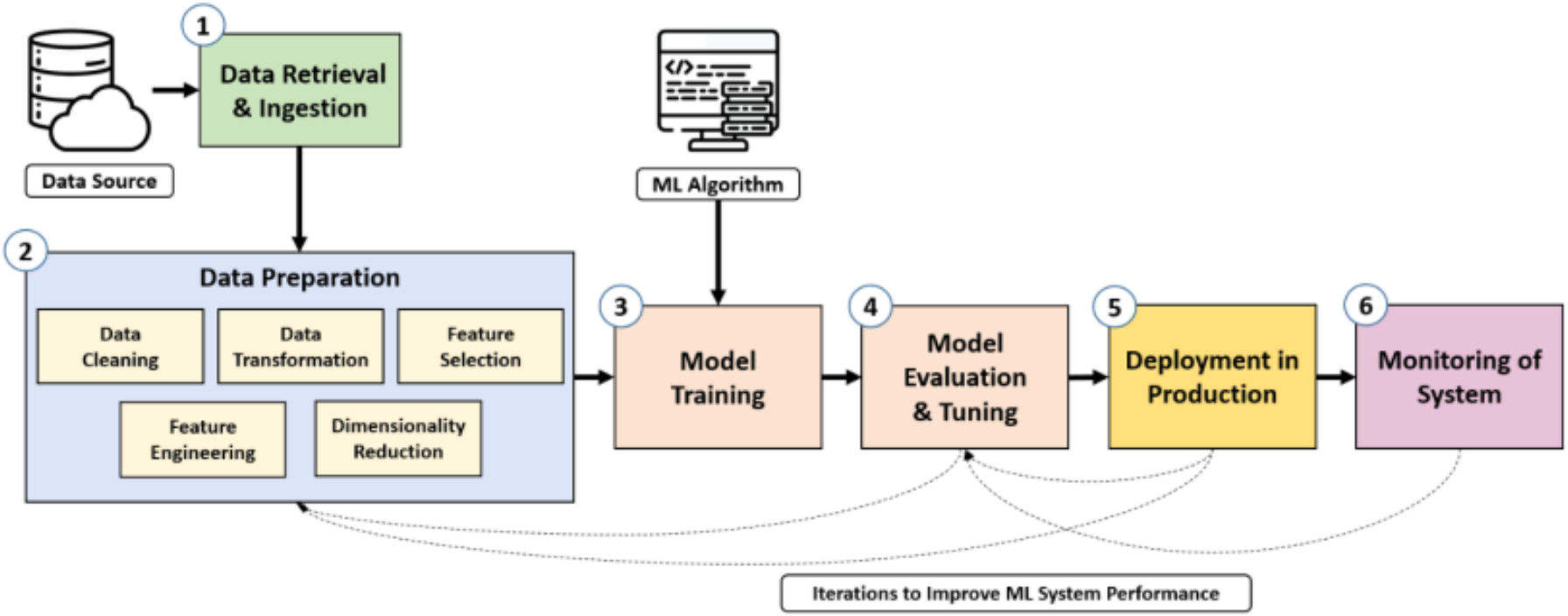
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Data pipelines



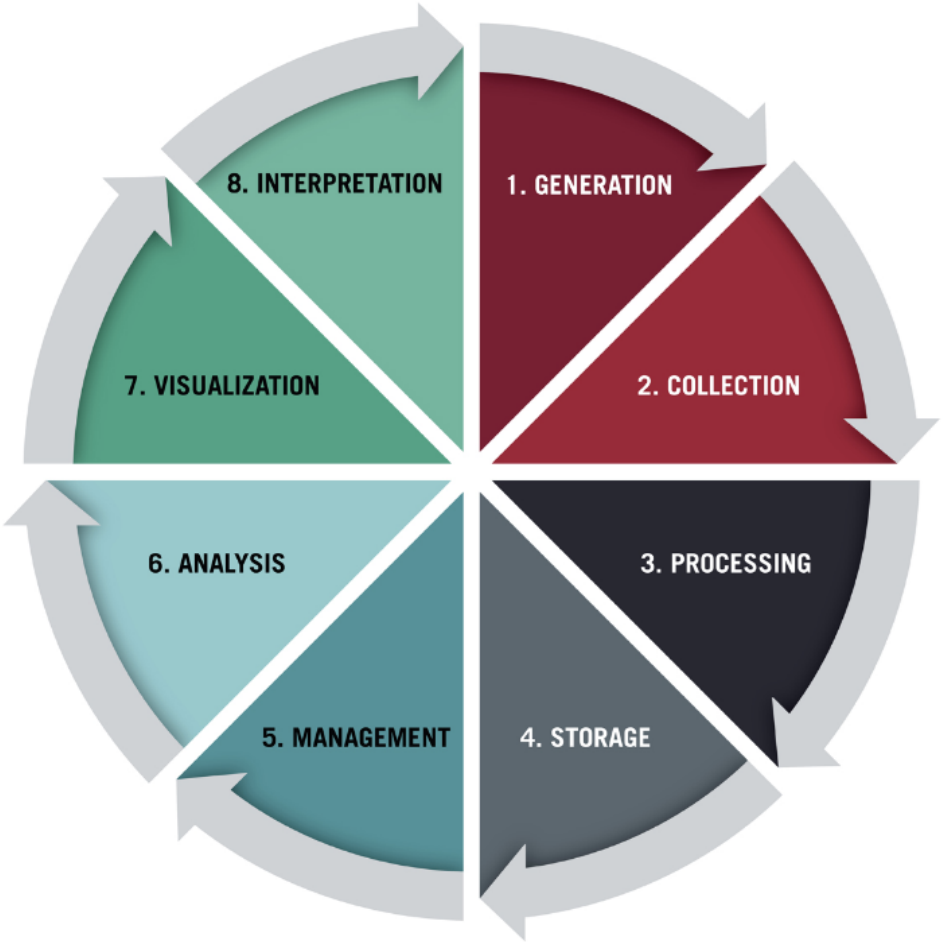
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Data pipelines



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Data lifecycle



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Data engineering activities

- Most data scientists spend **20 percent** of their time **building data models** and **analyzing model results**. What do they do with the remaining **80 percent** of their time? **Data engineering**!
- Data engineering refers to **designing** and **building systems** for **collecting** and **storing** at **scale**.
- Data engineering is the practice of **data processing**, **cleaning**, and **preparing ready-to-use data** for subsequent analysis, data science, and AI implementation; which often involves ML.

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Data eng. activities

- **Data acquisition/ingestion/extraction**: Data engineers develop pipelines to continuously **collect data** from a **variety of sources**—ranging from transactional databases, APIs, streaming data from sensors, logs, or external datasets. The goal is to **ingest** both **structured** and **unstructured data reliably**.
- **Data integration/loading**: These pipelines are designed to bring together data from disparate systems. Integration often involves **consolidating data** into a **central repository** (target systems) such as a **data mart** or **data warehouse**, where analysts and business intelligence tools can easily access and query the information.

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Data eng. activities

- **Data processing pipeline**: Depending on the use case, pipelines may be designed for **real-time (streaming)** data processing or **scheduled batch processing**. Real-time systems are critical for applications like fraud detection or live analytics, while batch processes might handle large volumes of data on a regular schedule.
- **Data storage**: Data engineers select and design appropriate storage solutions that can **handle large volumes of data efficiently**. This includes designing data lakes (for raw, unstructured data) and data warehouses (for processed, structured data).

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Data eng. activities

- **Data processing architecture**: Modern data systems must scale horizontally or vertically to accommodate growing data volumes and user demands. This involves choosing the right architecture—whether it's a **cloud-based system** or **on-premise solution**—and configuring it to optimize performance.
- **Data parallel processing**: Given the massive amounts of data and the need for rapid processing, data engineers often work with **distributed computing** frameworks (like Apache Hadoop or Spark) that allow parallel processing across clusters of machine.

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Data eng. activities

- **Data transformation**: Raw data is rarely in a format suitable for analysis. Transformation involves **normalizing** and **converting** the data into a standardized format. This might include data enrichment, aggregation, or filtering out irrelevant data.
- **Data validation & cleaning**: Ensuring data accuracy and consistency is a central activity. Data engineers implement automated validation checks to identify and correct errors, **remove duplicates**, and **handle outliers & missing values**.
- **Data catalog and metadata**: creating, maintaining, and utilizing an organized inventory of data assets and their associated context.

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Data eng. activities

- **Data monitoring**: Building resilient systems means setting up monitoring and logging mechanisms that **detect issues in real time.** Data engineers design pipelines with **failover** mechanisms, **retries**, and **alerts** to address errors and maintain system reliability.
- **Performance optimization**: regular performance **tuning of queries**, data retrieval processes, and storage solutions is essential. This can involve **indexing**, **partitioning** data, and **optimizing resource allocation** to ensure that data systems perform efficiently even under heavy loads.

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Data eng. activities

- **Data sharing and security**: While making data **accessible and available** to different stakeholders, in the same time data engineers implement robust security measures such as **encryption**, **access controls**, and **secure data transfer** protocols to **protect sensitive information (personal data protection** using **Privacy Enhancing Technologies)**.
- **Data compliance and governance**: With regulations like **GDPR, HIPAA, and CCPA**, ensuring that data handling processes comply with legal requirements is critical. This involves setting up **policies** for data retention, preservation, auditing, and monitoring data usage.

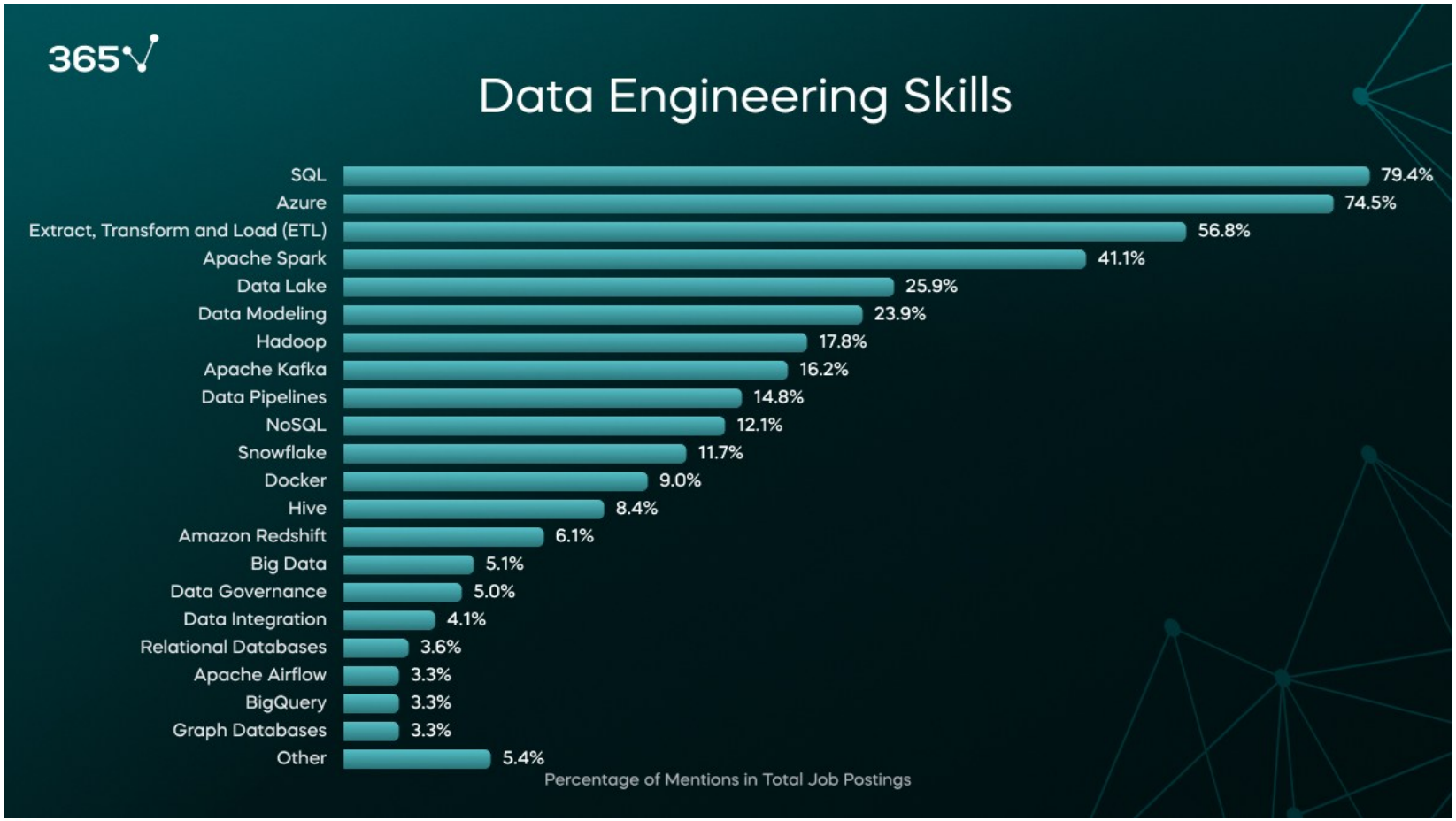
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Data engineering activities

- **Tooling & system management**:
  - **Database technologies**: Data engineers work with various database systems—**SQL** for structured data and **NoSQL** for flexible, unstructured data storage.
  - **Big data frameworks**: Tools such as Apache **Hadoop**, **Spark**, and **Kafka** are central to handling big data.
  - **Cloud platforms & infrastructure automation**: Many modern data infrastructures are hosted on cloud platforms like **AWS**, **Azure**, or **Google Cloud**.
  - **Workflow orchestration**: Scheduling and managing complex data workflows require orchestration tools such as **Apache Airflow**.

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Data engineers' skills



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Data engineers' certifications

- https://www.cloudera.com/services-and-support/training/cdhhdp-certification/cca-spark.htm
- https://learn.snowflake.com/en/certifications
- https://www.databricks.com/learn/training/certification#data-engineer
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